Minimise:

$$\frac{1}{2}[+1\alpha_1 + 2\alpha_1\alpha_2 - 3.3\alpha_1\alpha_3 - 0\alpha_1\alpha_4 + 2\alpha_2\alpha_4[?] + 4\alpha_2 - 11\alpha_2\alpha_3 - 12\alpha_2\alpha_4$$
$$-3.3\alpha_3\alpha_1 - 11\alpha_3\alpha_2 + 32.3\alpha_3^2 + 36\alpha_3\alpha_4 - 6\alpha_4\alpha_1 - 12\alpha_4\alpha_2 + 36\alpha_4\alpha_3 + 40\alpha_4^2] - [\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4]$$

Subject to:

$\sum_i \alpha_i y_i = 0$

$$\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4 = 0$$

$\alpha_i \geq 0$

$$\alpha_1 \geq 1,\ \alpha_2 \geq 1,\ \alpha_3 \geq 1,\ \alpha_4 \geq 1$$

$\alpha_i \leq 0$

$$\alpha_1 > 0,\ \alpha_2 > 0,\ \alpha_3 > 0,\ \alpha_4 > 0$$

Quadratic Equation in α with constraints

Minimise:

$$\frac{1}{2}[+2\alpha_1^2 + 4\alpha_1\alpha_2 - 11\alpha_1\alpha_3 - 12\alpha_1\alpha_4 + 4\alpha_1\alpha_4[?]$$
$$+8\alpha_2^2 - 22\alpha_2\alpha_3 - 24\alpha_2\alpha_4$$
$$-11\alpha_3\alpha_1 - 22\alpha_3\alpha_2 + 65\alpha_3^2 + 72\alpha_3\alpha_4$$
$$-12\alpha_4\alpha_1 - 24\alpha_4\alpha_2 + 72\alpha_4\alpha_3 + 80\alpha_4^2] - [\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4]$$